

Whiteboards

Sat & Sun 8 PM to 11 PM

23-Apr-2022 to 15-May-2022

Best practices - modern automation

Compliance
Rule & Regulations

★ ★ ★ ★ ★

lots learning

DevSecOps

DevOps

AWS Cloud

Azure

GCP...

Computer Science

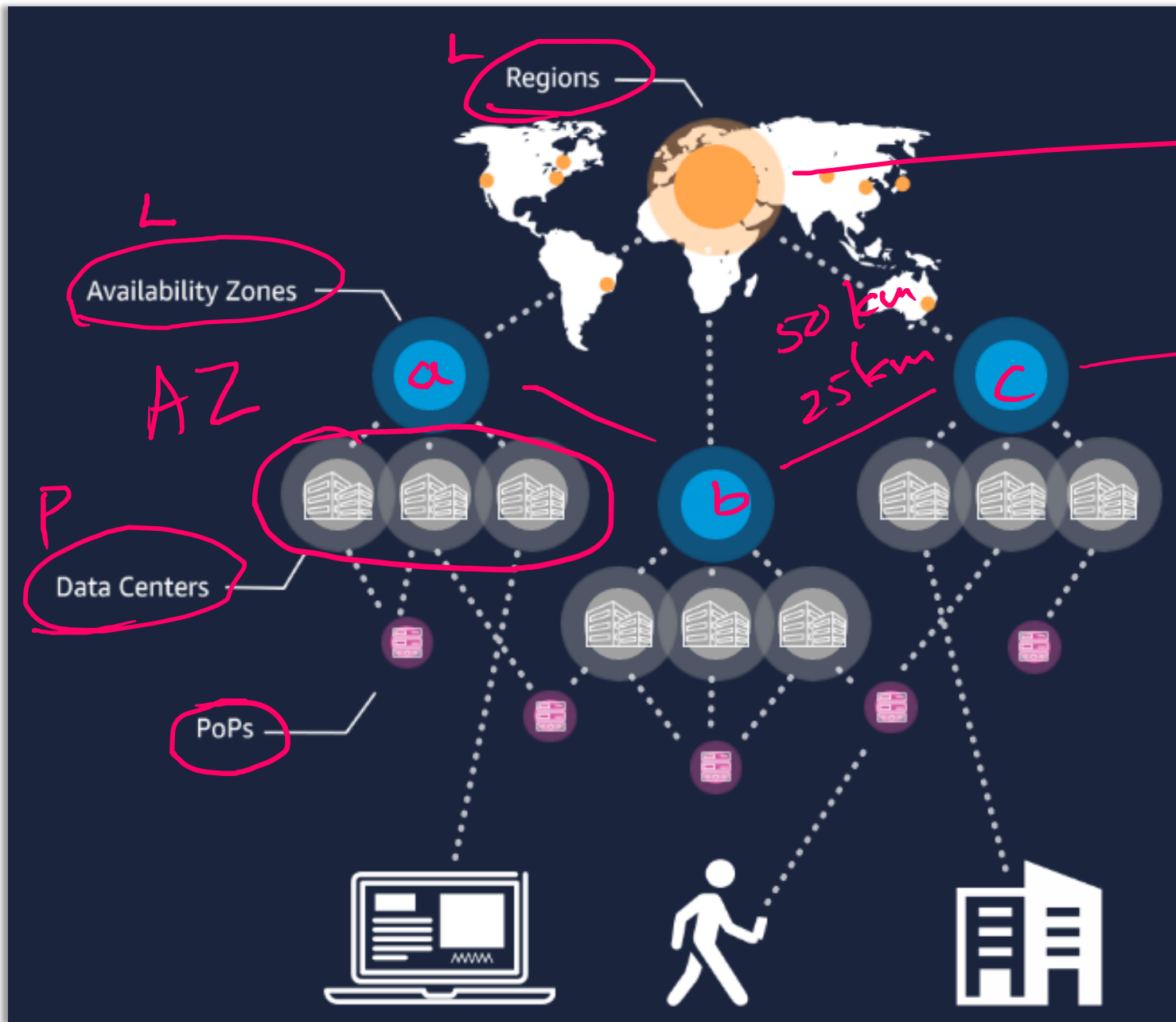
Programming

s/w testing

Architecture

Bits / Bytes

Mechanical
Electrical
Civie...



logical term
ap-south-1

logical term
ap-south-1a
-1b
-1c

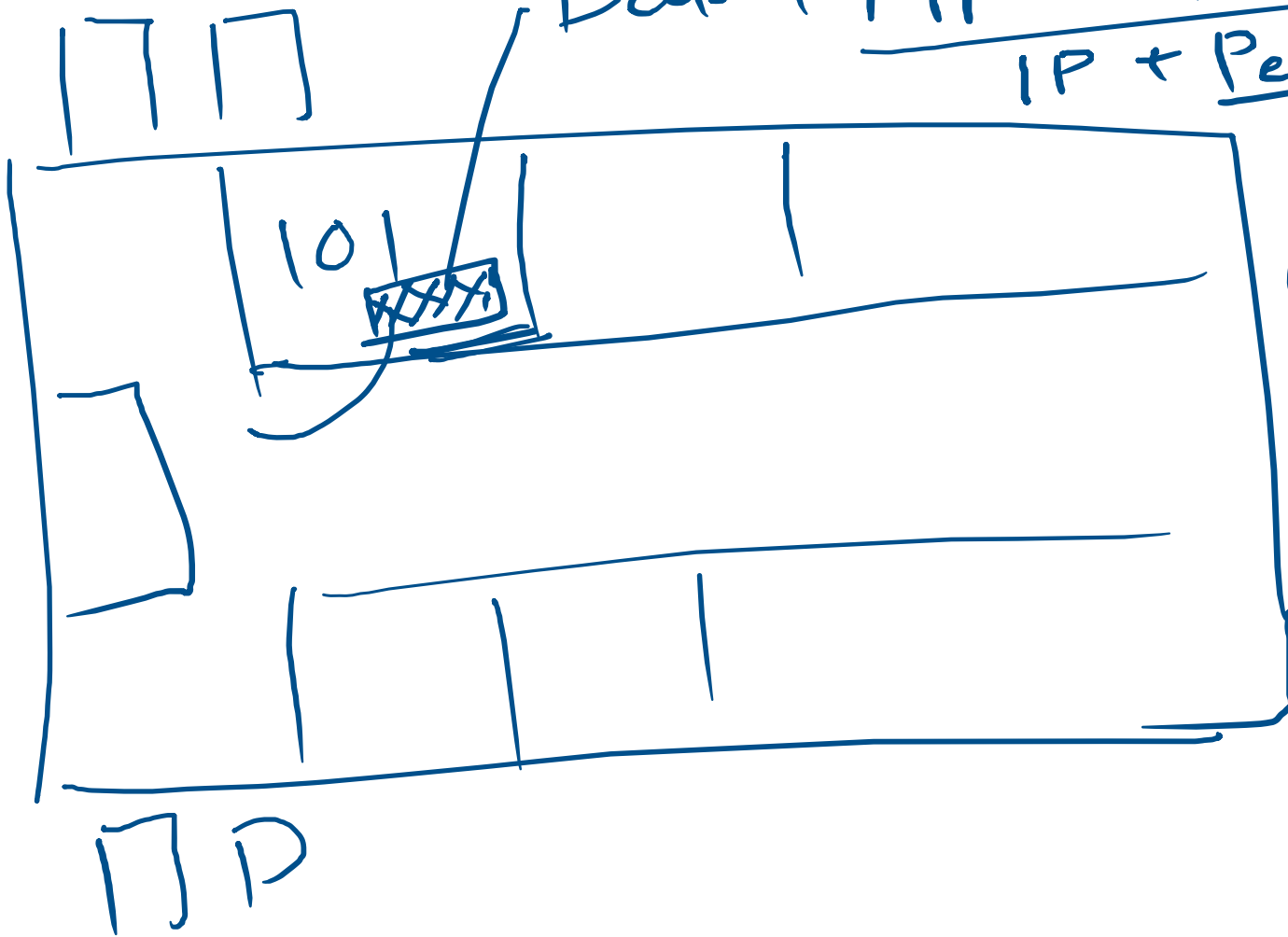
Local Zone
Wavelength Zone

Security
Water
Power
Internet
Consumables
...

Data + Application
IP + Pet

Cloud
Multitenant

Crowd

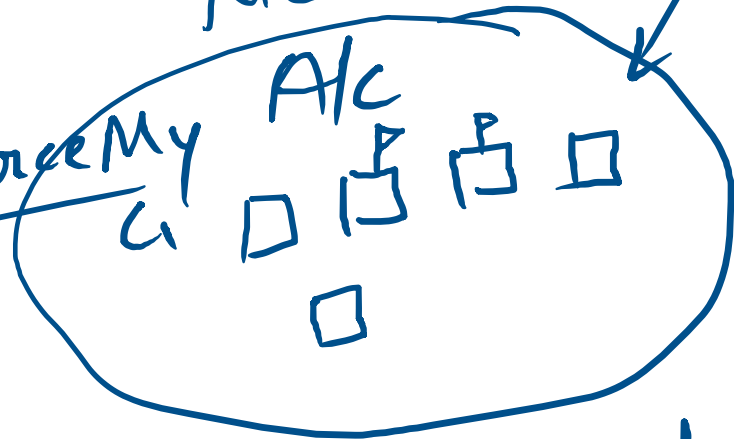


Crowd $\Rightarrow$ Multitenant

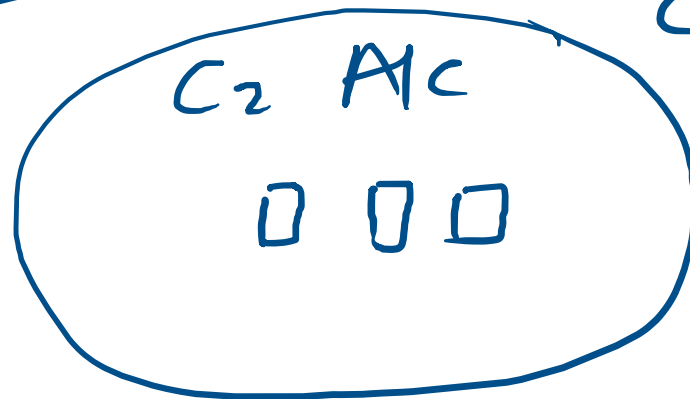
login
L VPC id

L resource My

Network

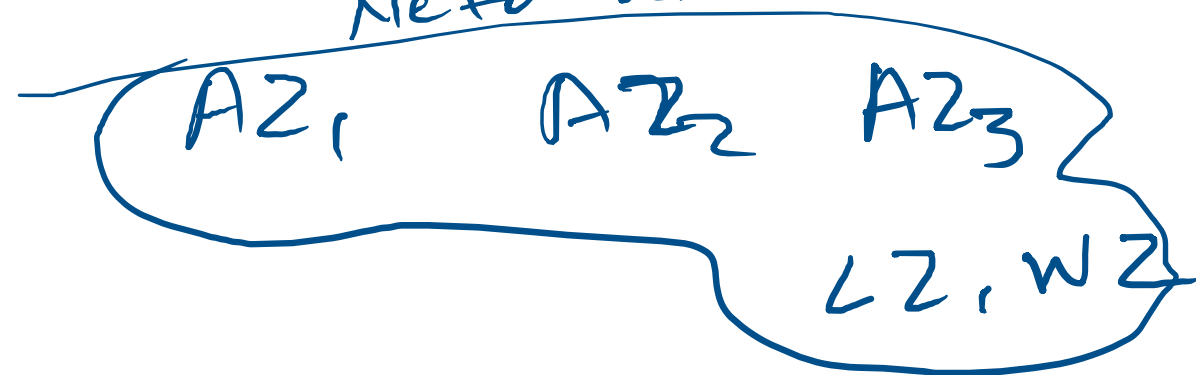


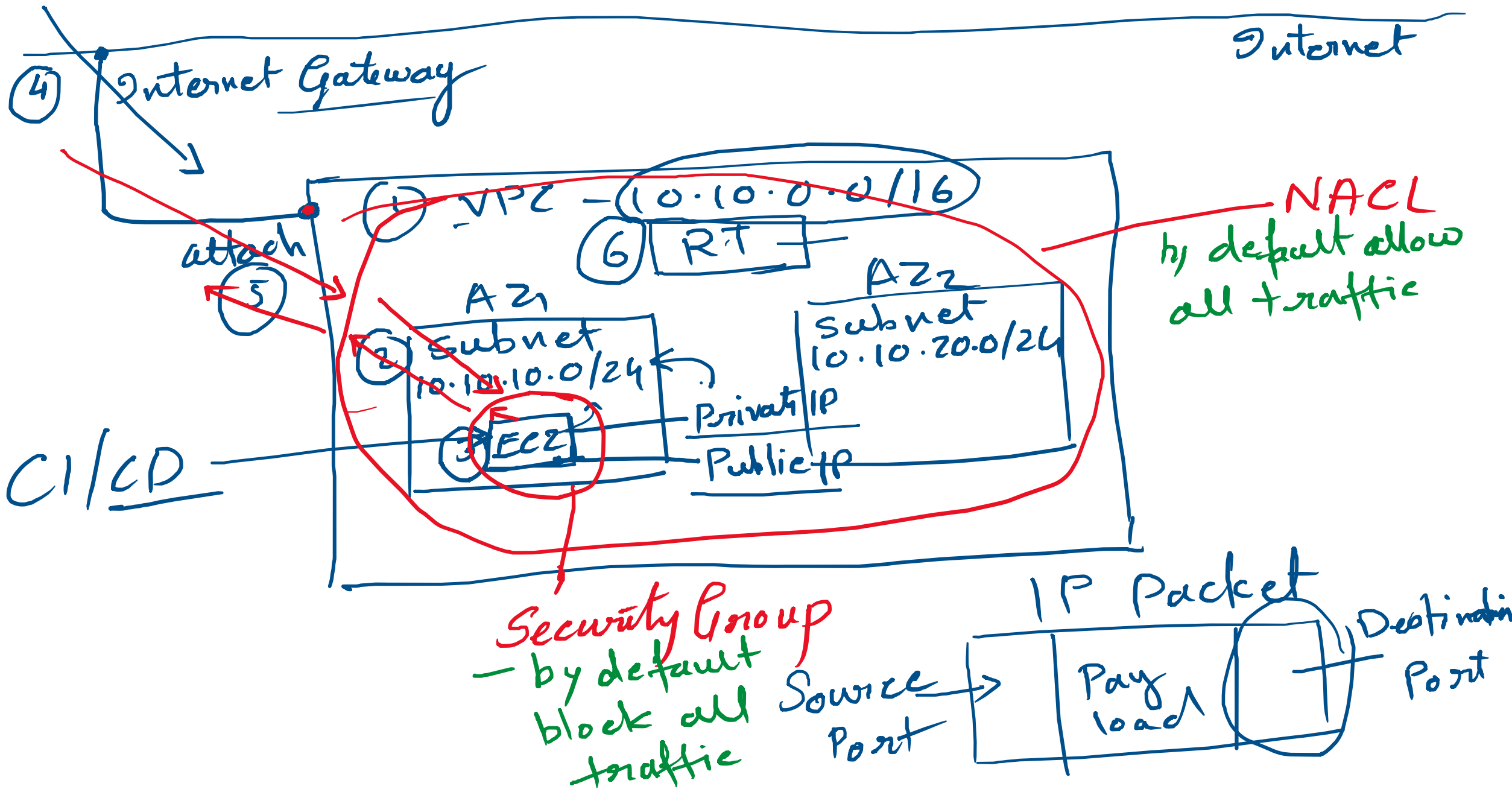
Isolation Boundary
VPC - Virtual Private Cloud

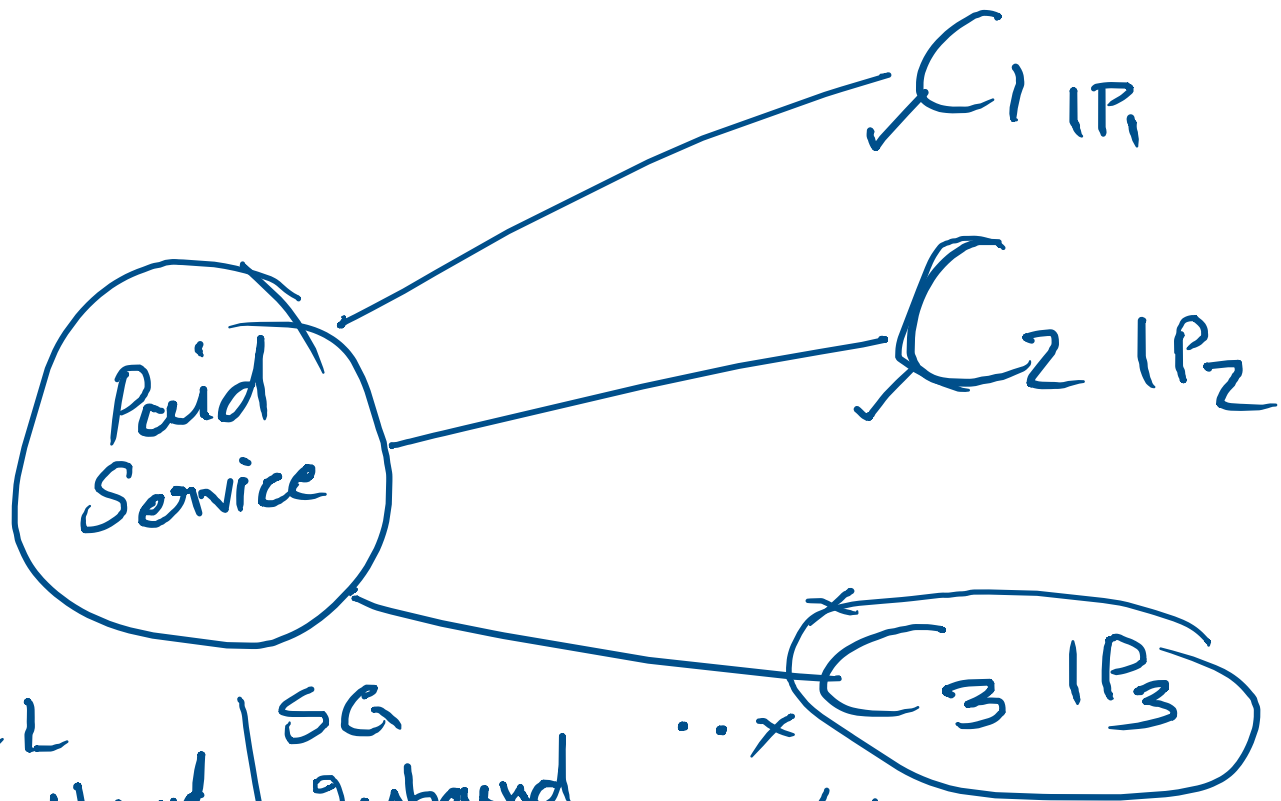


Virtual Machines

VPC - Regional
Network

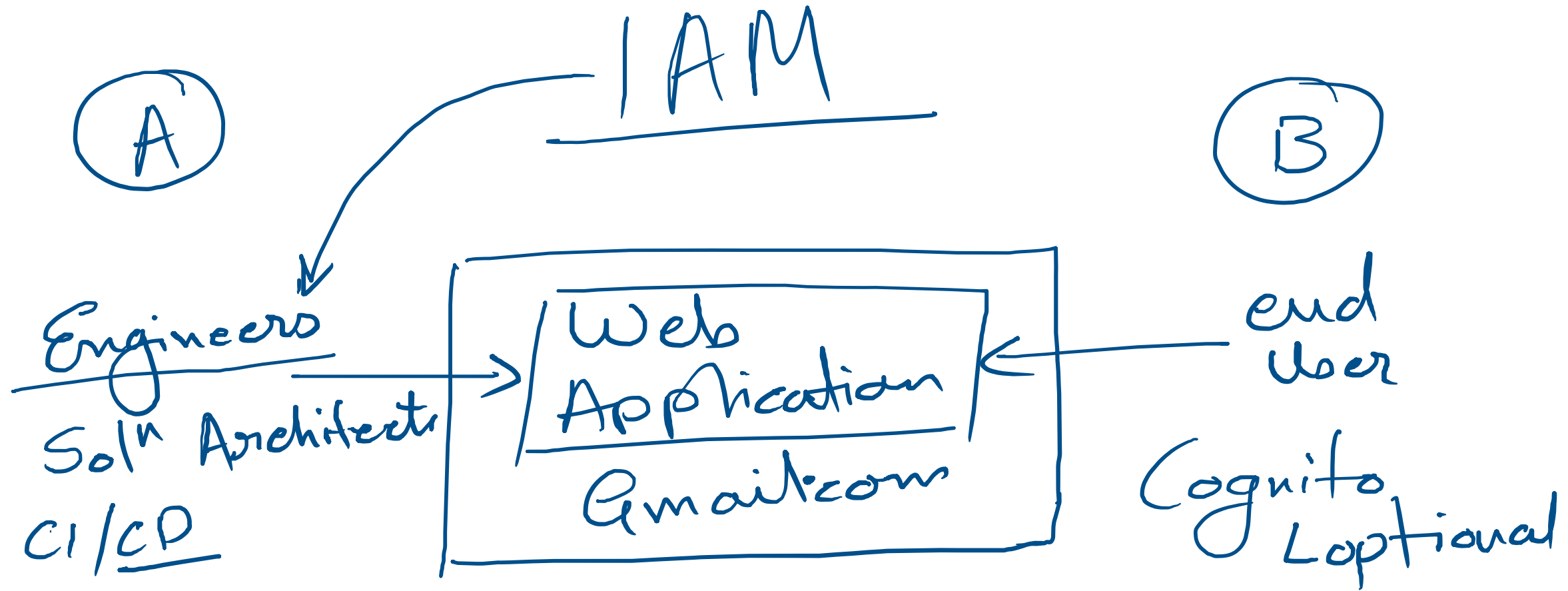


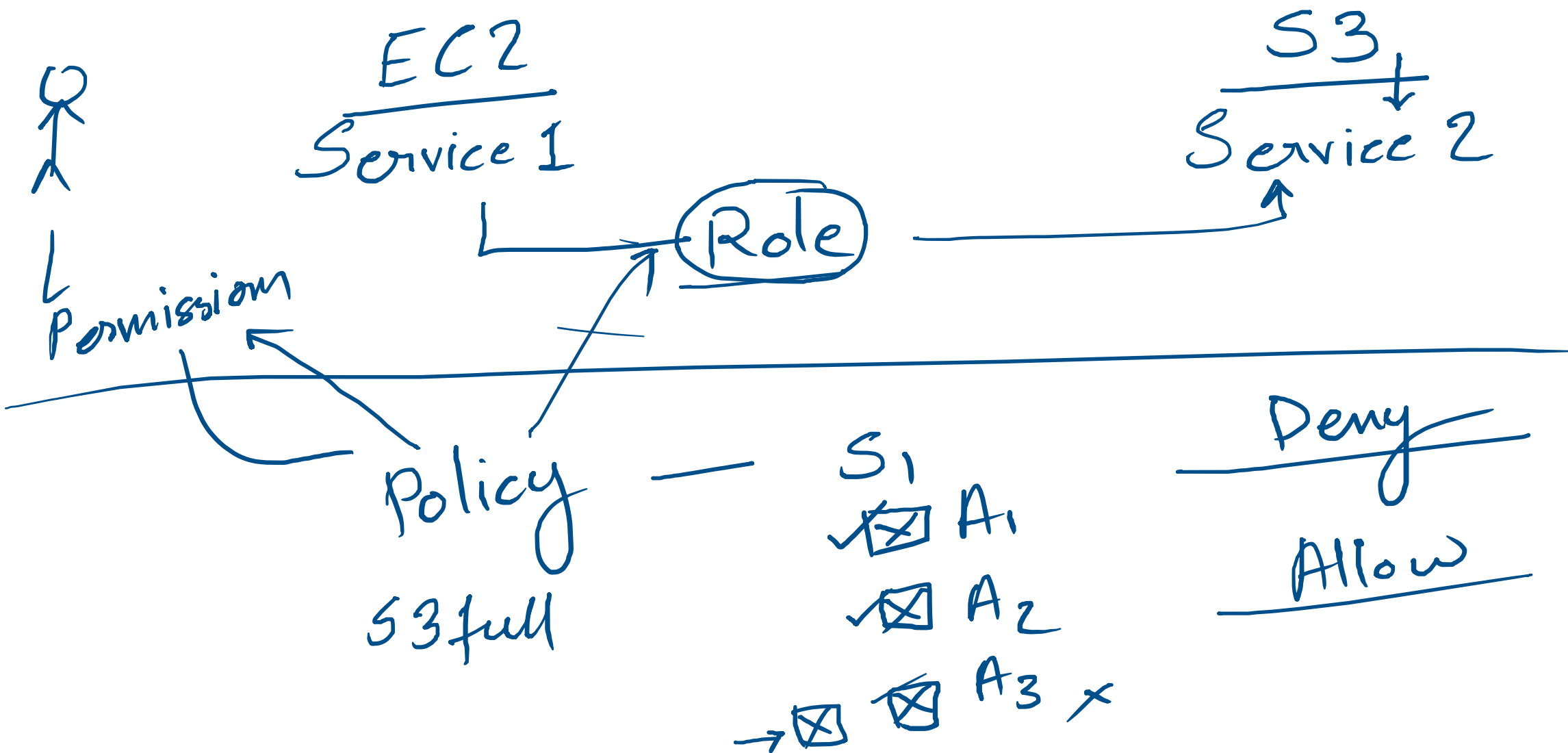




NACL
Inbound ✓ Outbound ✓

SG
Inbound





CI

DevOps

CD Deployment

DevSecOps

QA — UAT — PreP — Prod

CD Delivery

↳ to the biz

Code Deploy

Service
Role

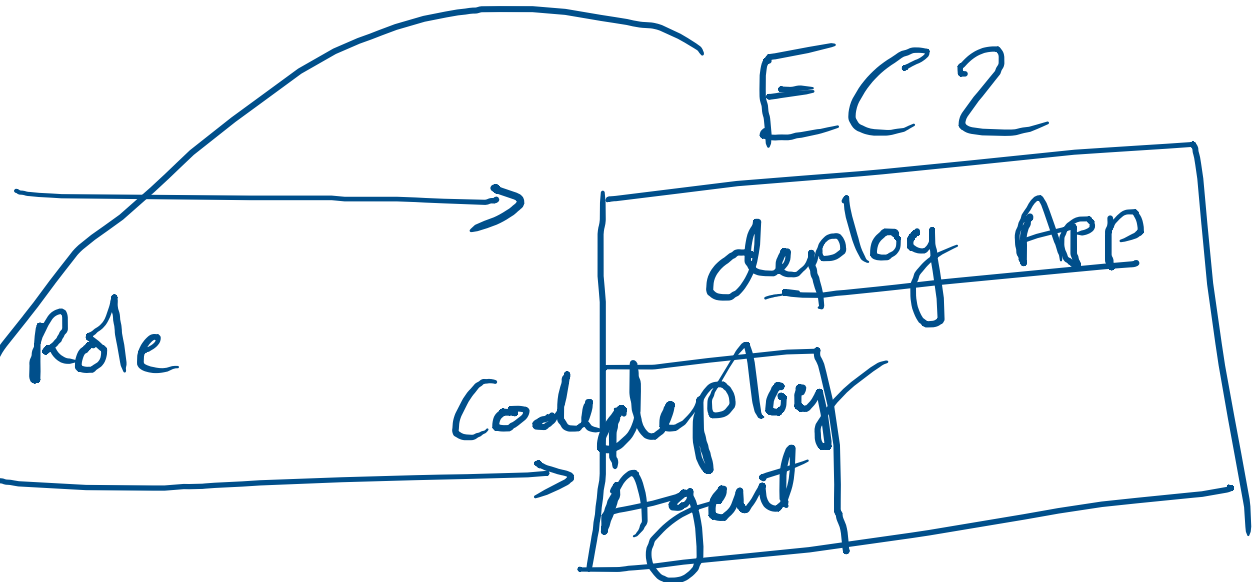
S3

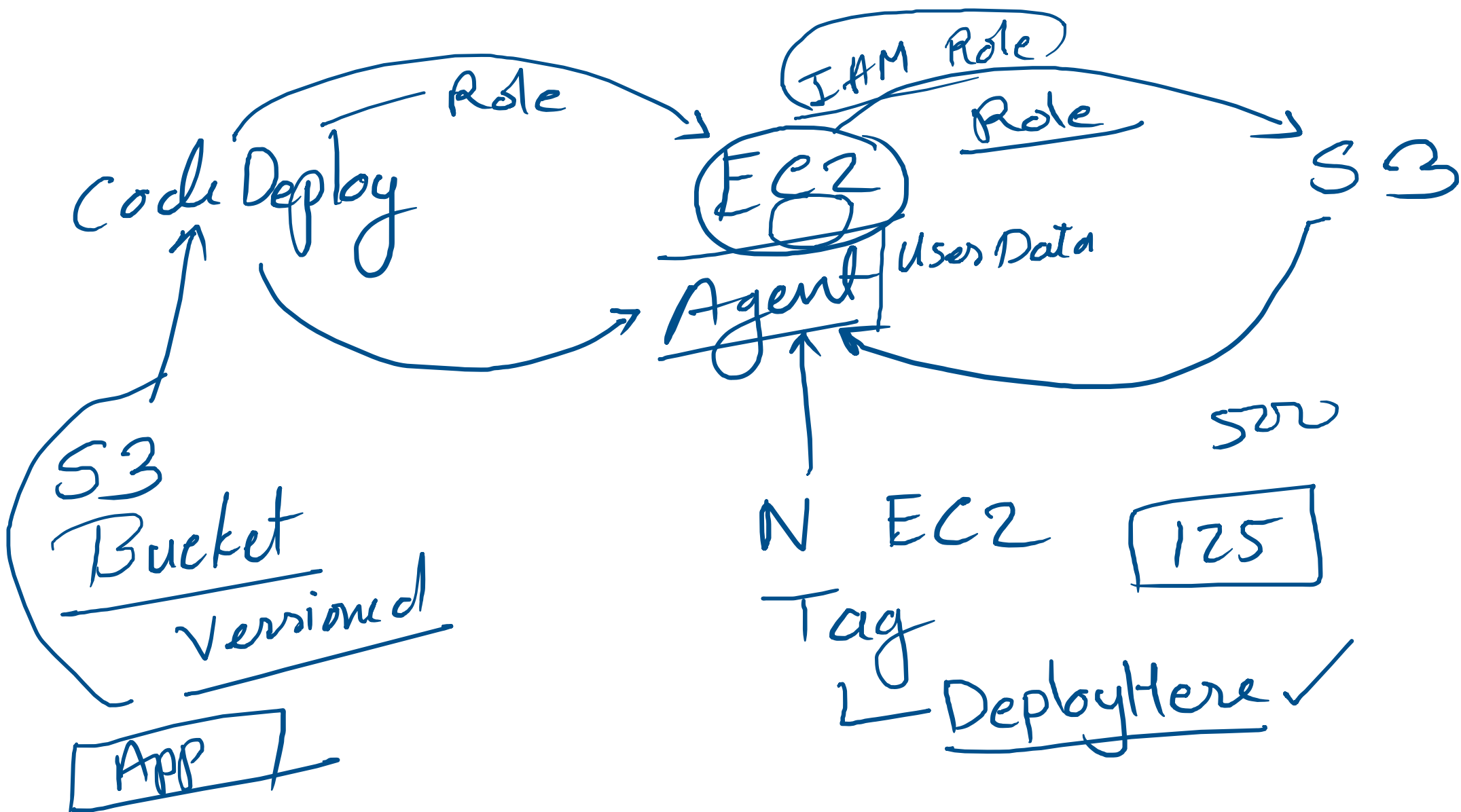
Role

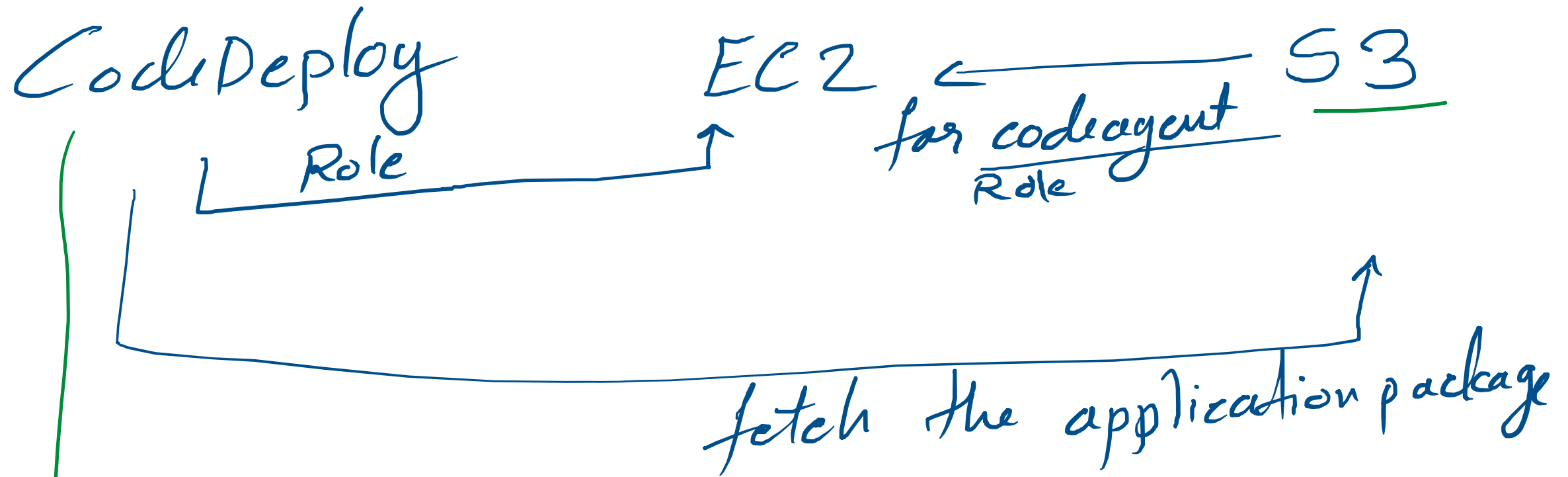
EC2

deploy App

CodeDeploy
Agent



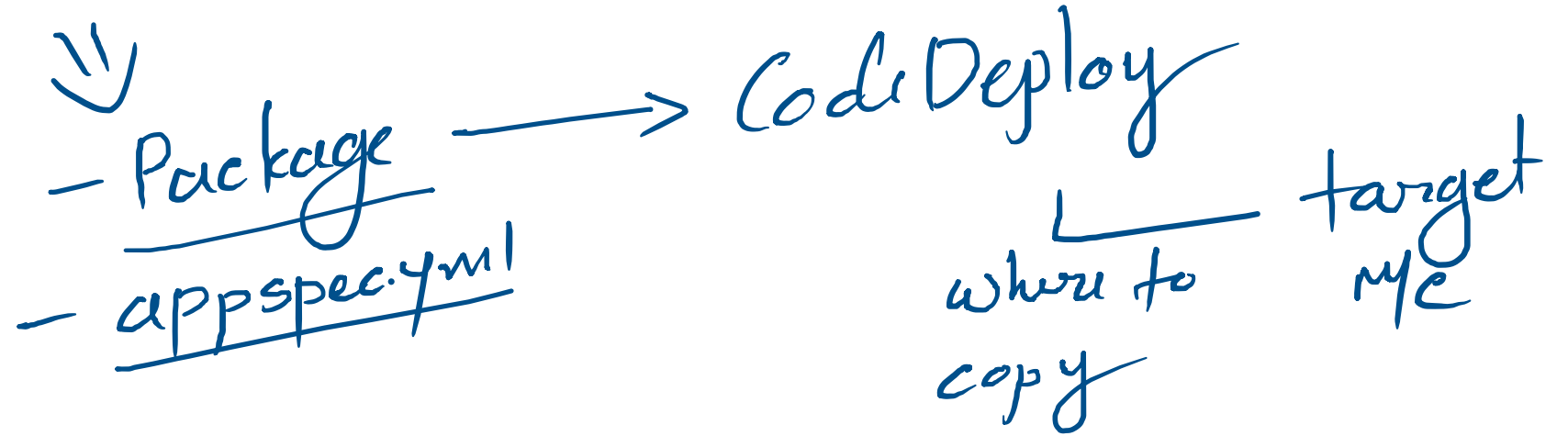
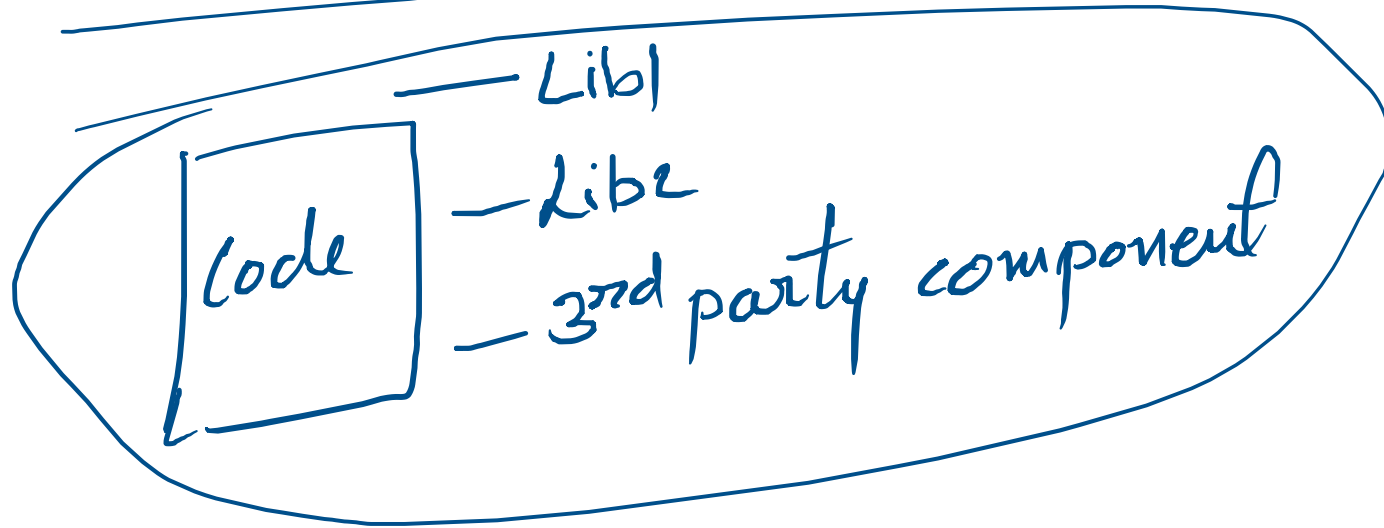




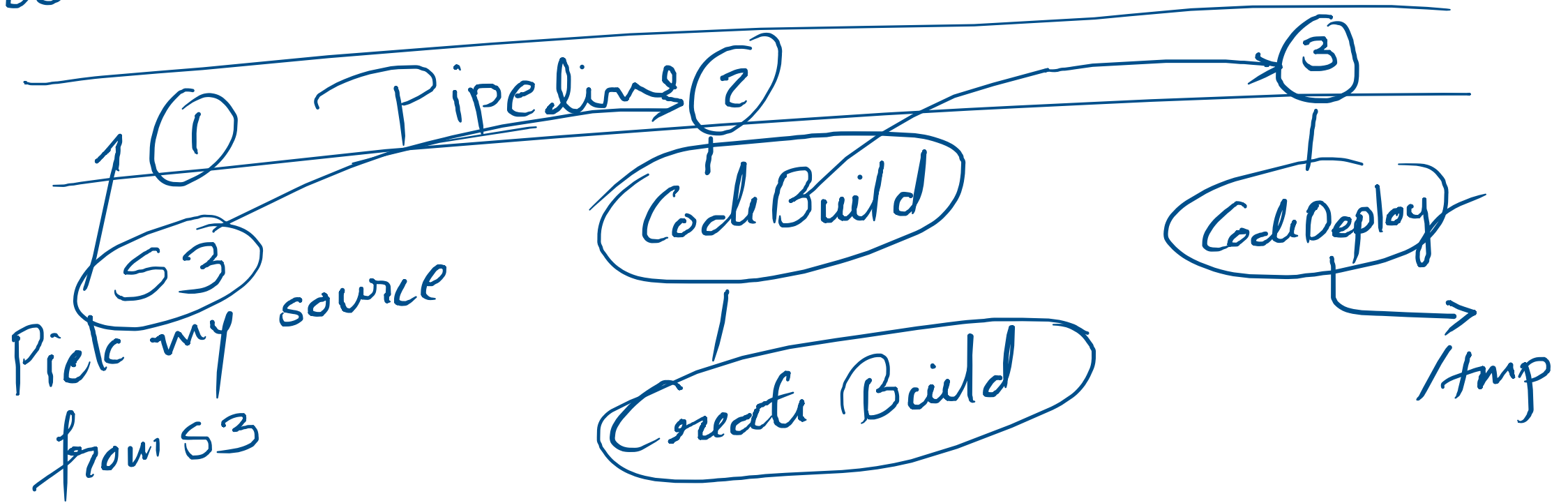
CodeDeploy Agent

- event — Download Bundle
- — ApplicationStop
 - — Before Install
 - Install

Code Build



Code commit



→ Big focus ← Employee

Code Warriors — VSS

Source control
code management
scm



12 hrs
Build
run test

smoke
black box
Regression

Deploy
email
Windows
10,000

Robohelp

QA

Category 1

Grammar

Category 2

Usability

Category 3

Unit
Integration

CD

UAT & Prod — CD

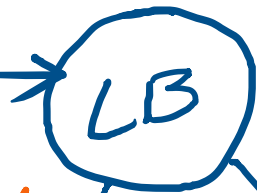
LB
HA
FT

Gmail $\Rightarrow$ 2.5 million
5 sec



URL: 7000

URL: 5000



endpoint
www.gmail.com

App-01



App-01



App-02

AZ₁



10.10.10.41

AZ₂



10.10.20.226

AZ₁

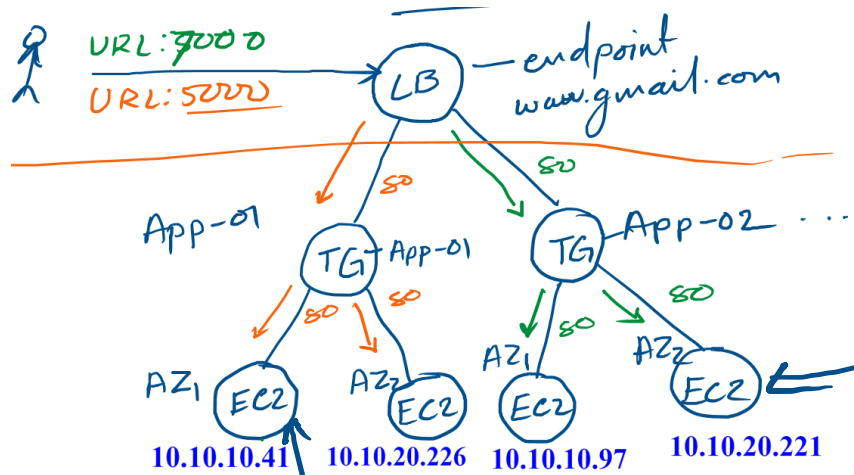


10.10.10.97

AZ₂



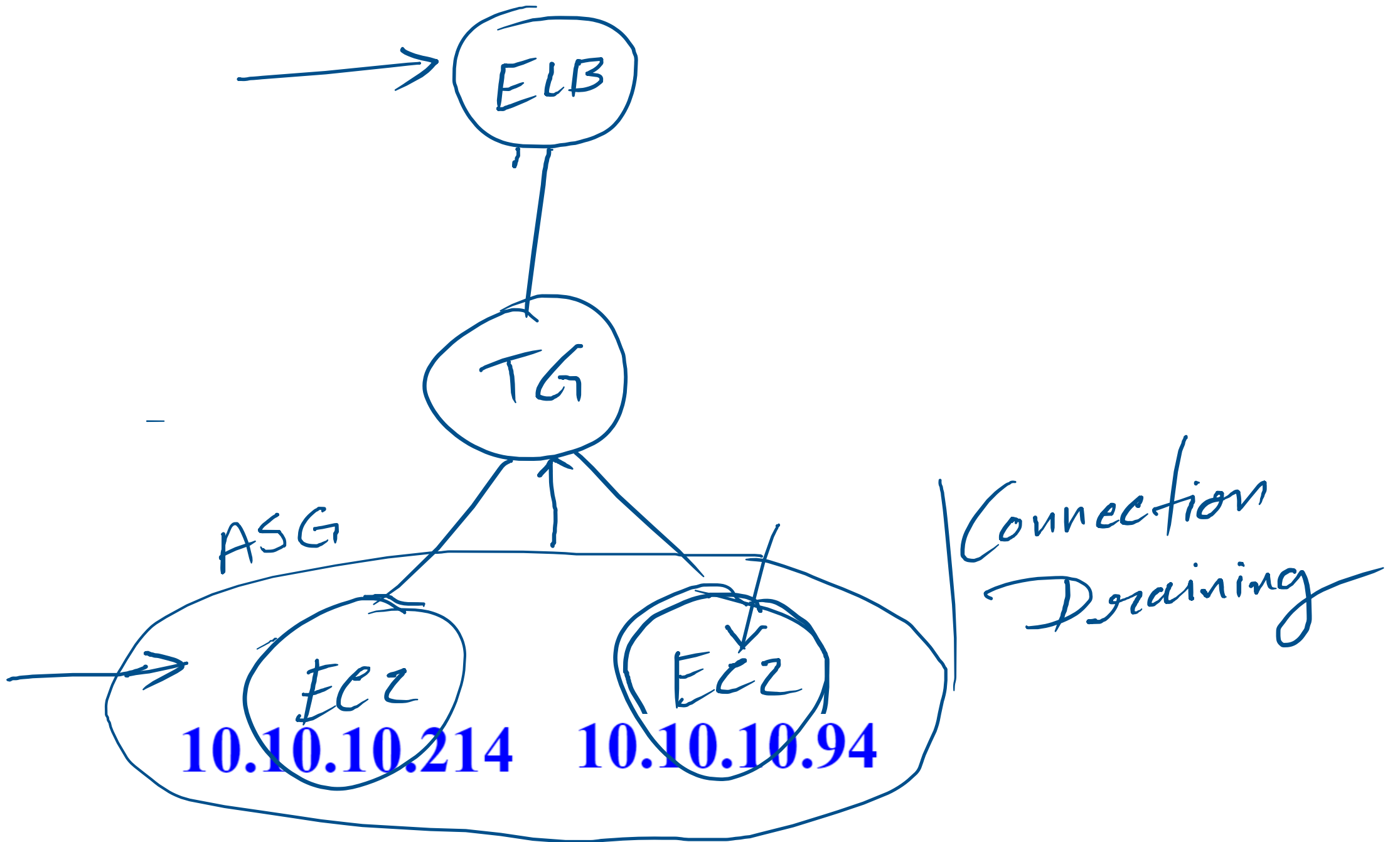
10.10.20.221



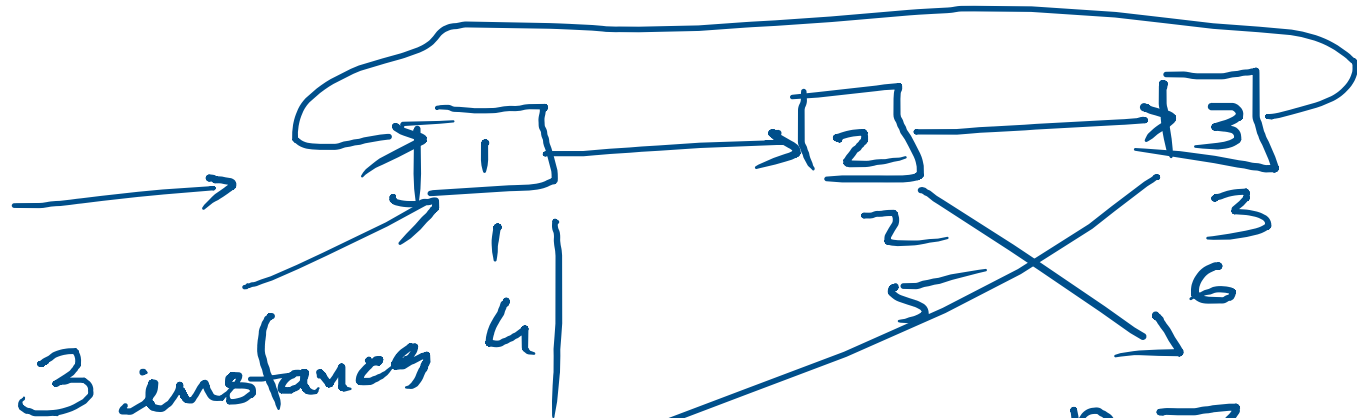
Deploy here

CodeDeploy must not deploy directly

1. De registers the instance from TG
2. Wait for the request instance may be serving to complete $\Rightarrow$ connection drain
3. deploy App
4. Register the instance with TG



LB ✓
HA ✓
FT



3 instances

AZ₁

AZ₂ horizontal out

→ Desired. 2

Min — 1
Max — 5



Ag CPU 70%
40%
Avg CPU < 20%

3 min SLA
sec

adding to the Group



AMI

Instance type

Key-Pair

Security Group

Public IP

User data

IAM Role

Launch
Template

v1

↓
v2

ASG

supports
inheritance

Launch
Configuration

old

2006 Google

Linux Containers

Expensive Server for multiple projects

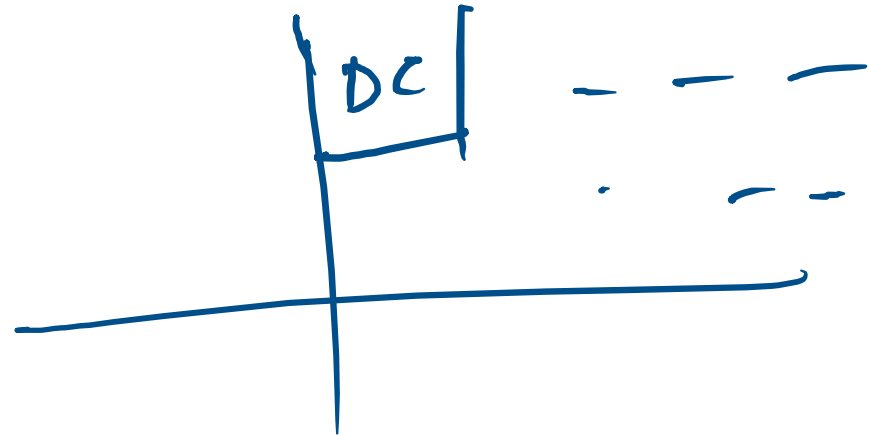
①

4 Core	32 GB
	512 GB

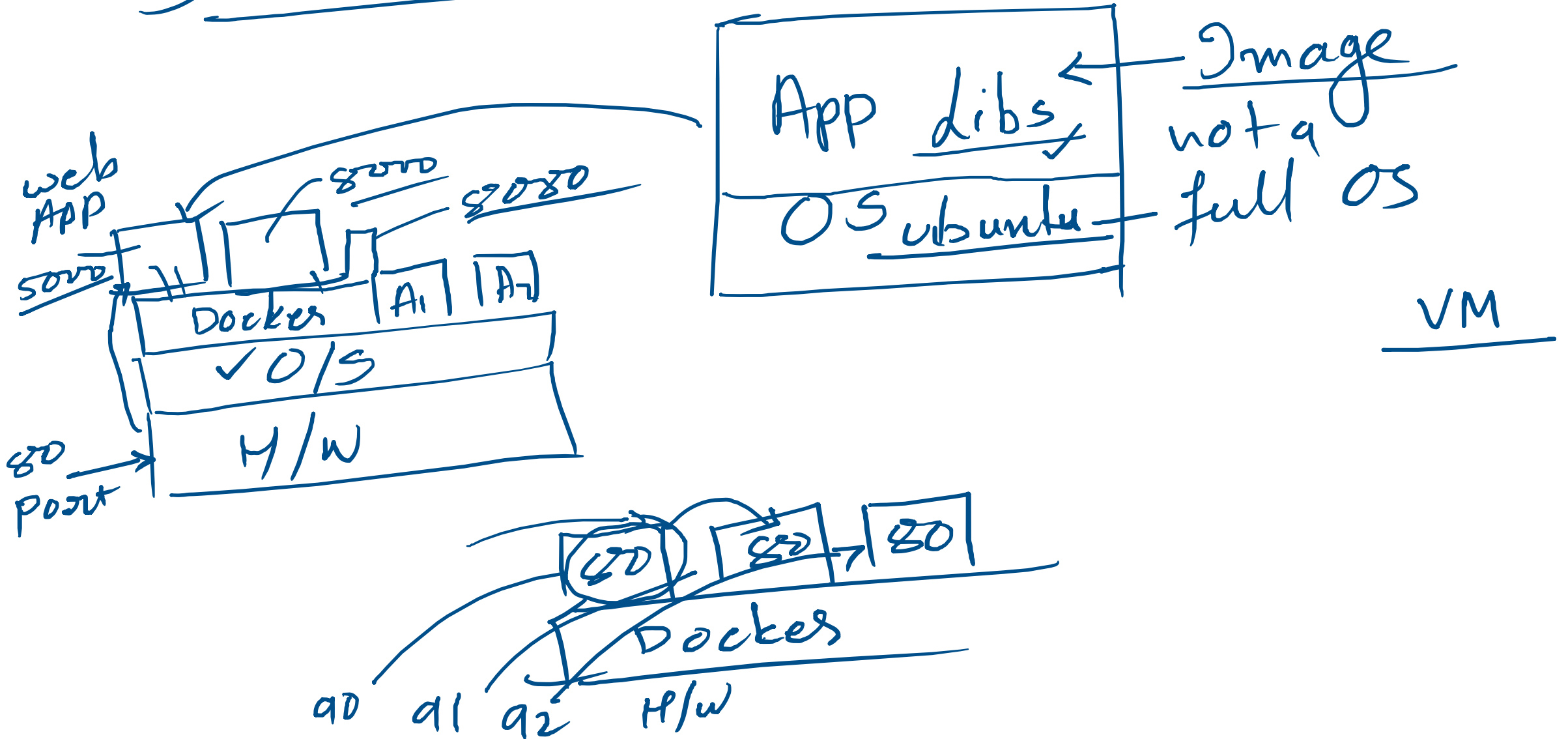
> CPU utilization

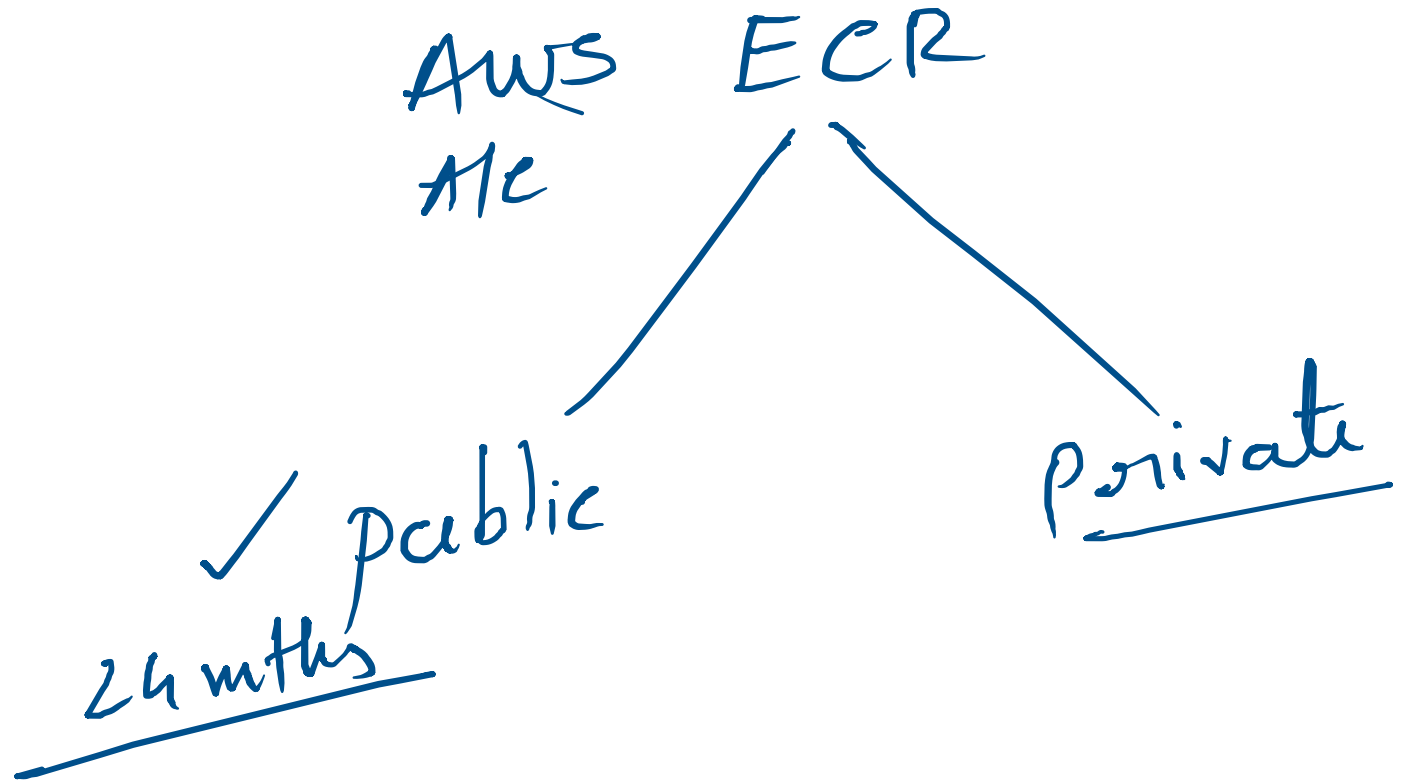
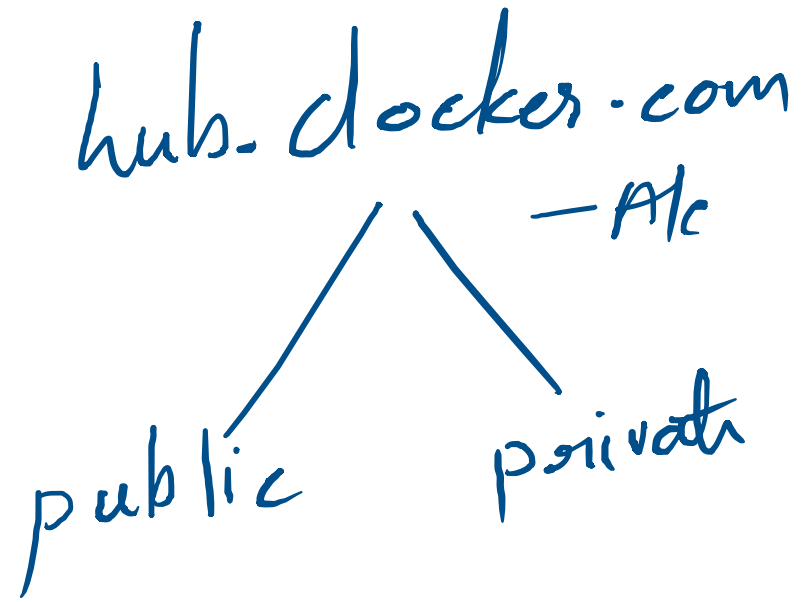
\$

dev		Operations	
Ⓐ	V ₂	→ A ₁	V ₁
L ₁	V _{1.2}	A ₂ ✓	L ₁ V _{1.0} ✓
L ₂	V _{2.4}	A ₃ ✓	L ₂ V _{2.0} ✓
		A ₄ ✓	



Docker 2011





ECS

Service

Emp details

Cluster

Cluster

Lservers

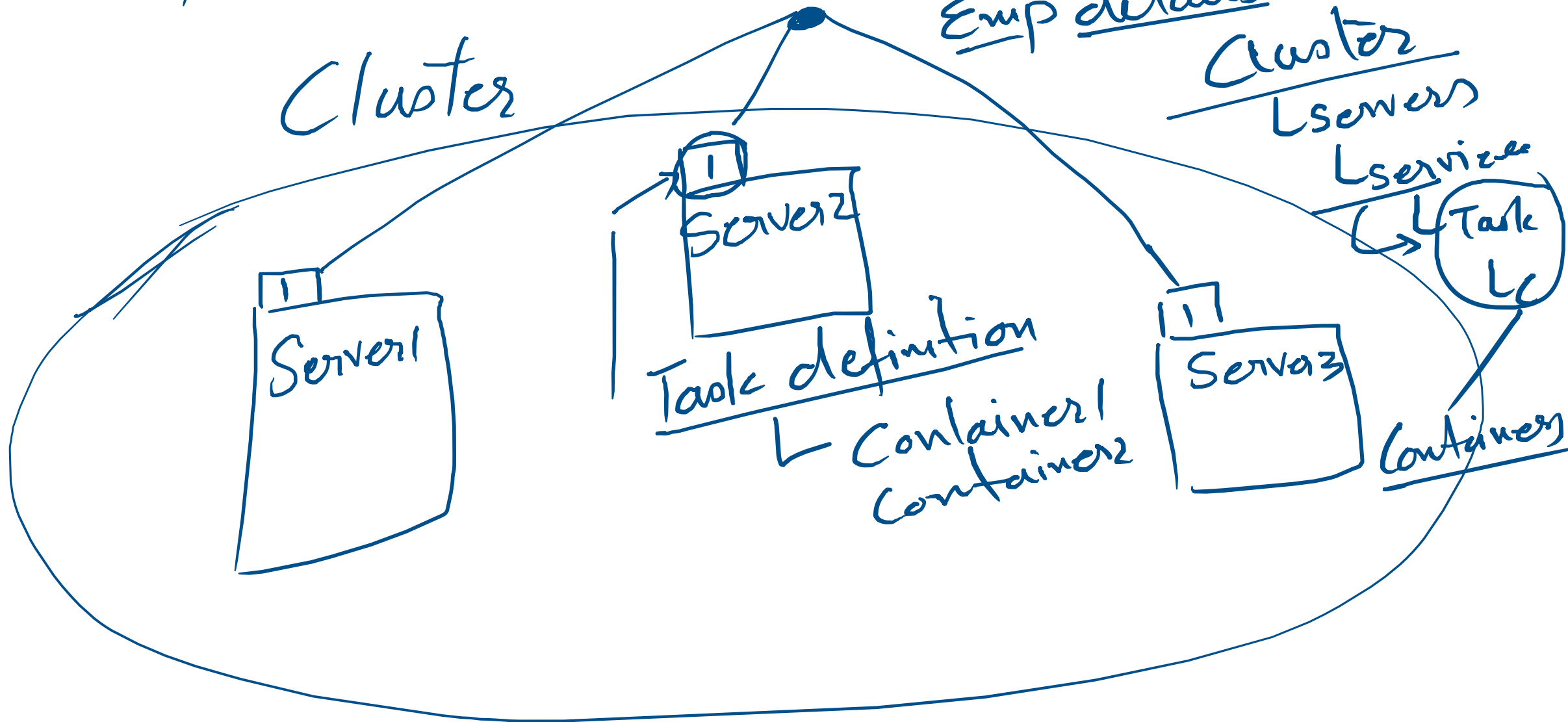
Lservice

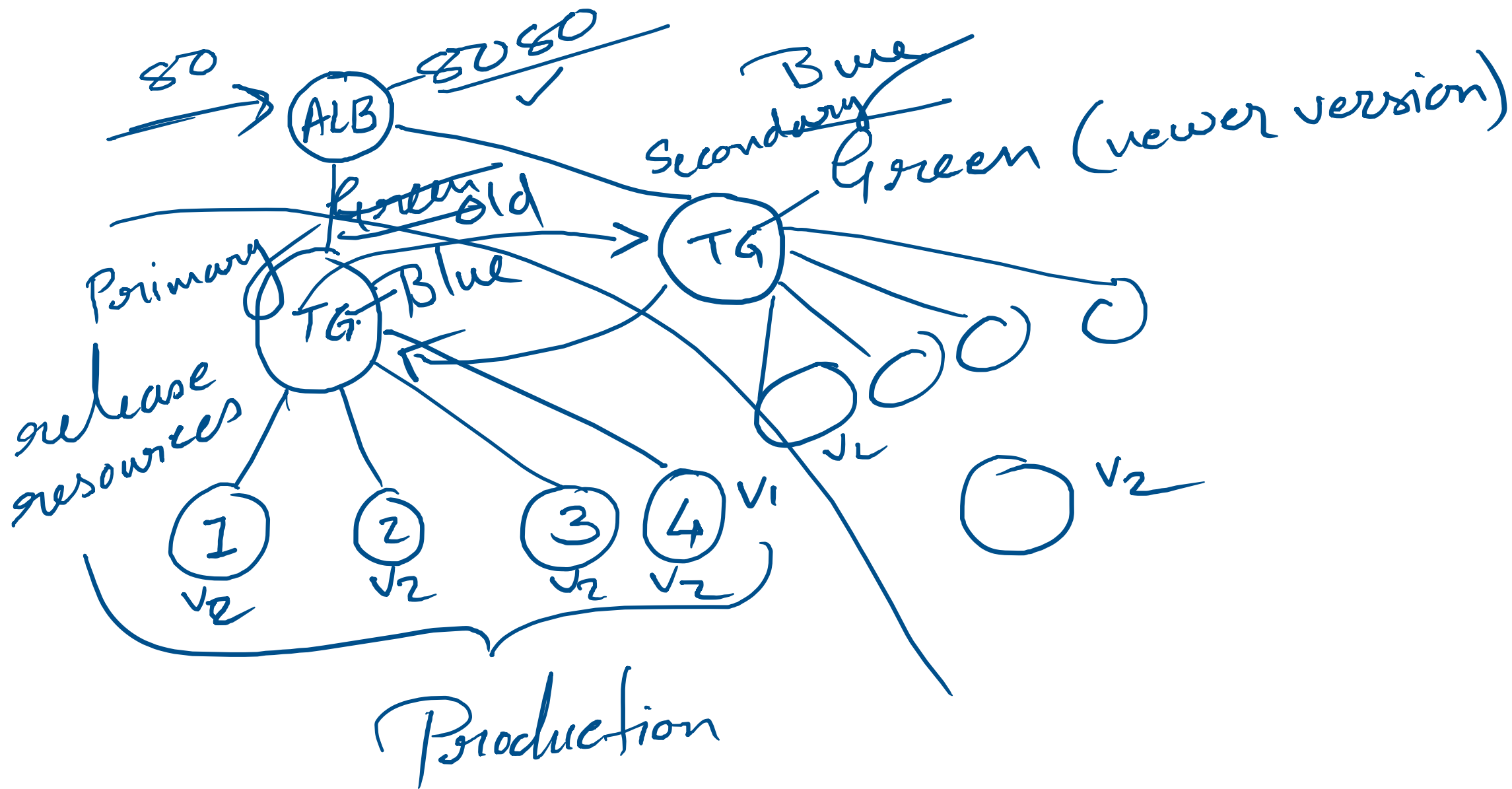
LTask

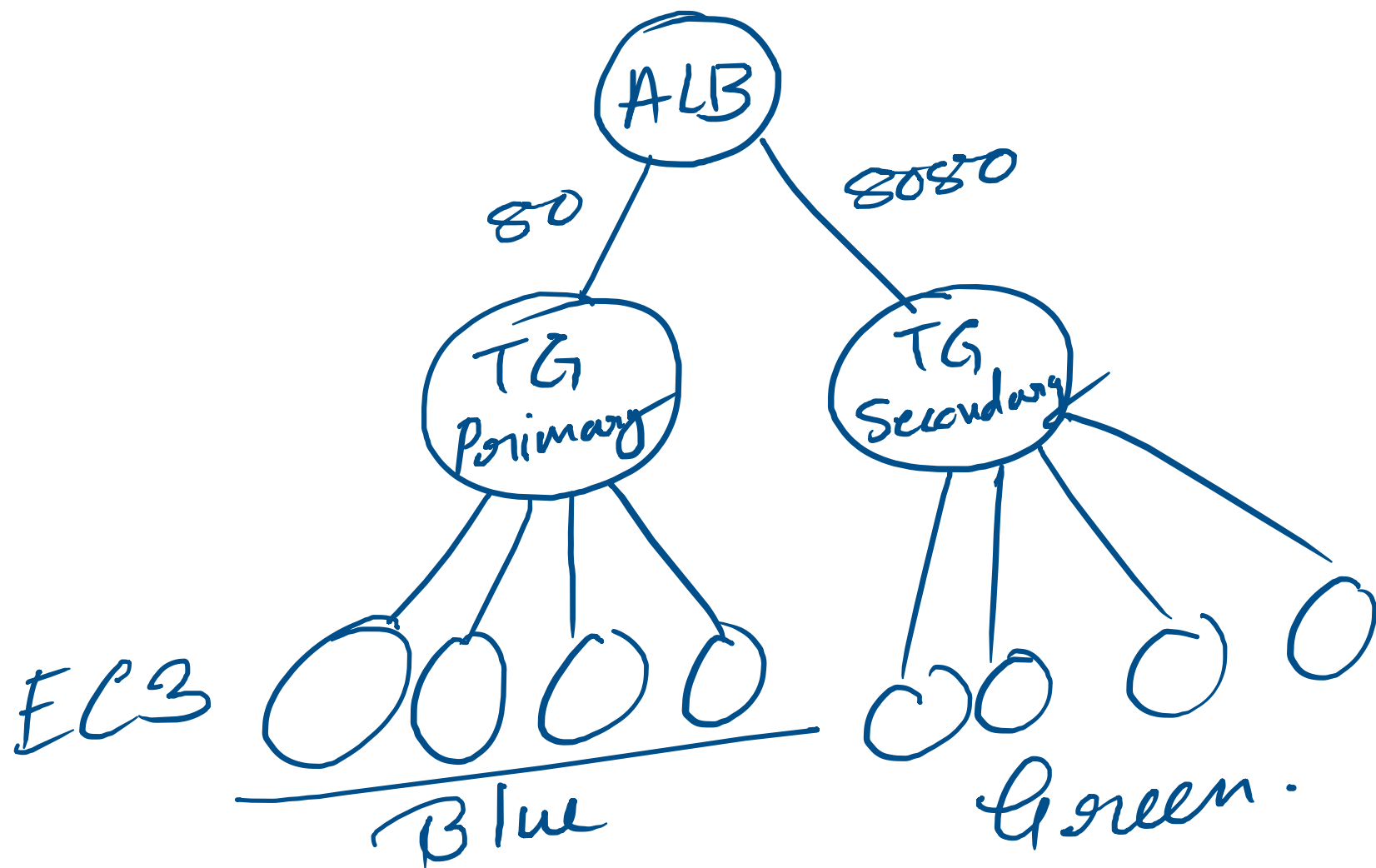
Task definition

L Container1
Container2

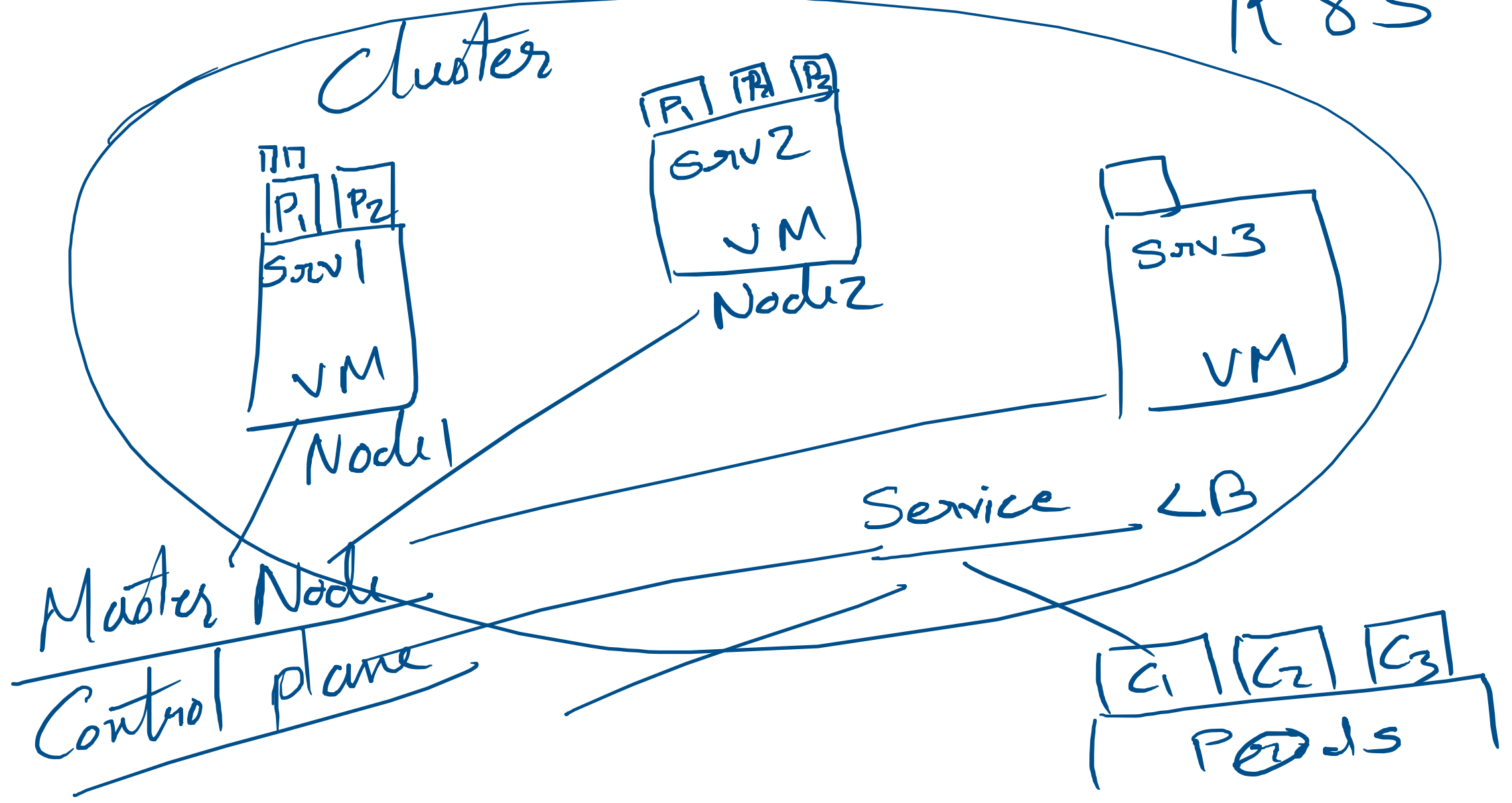
Containers

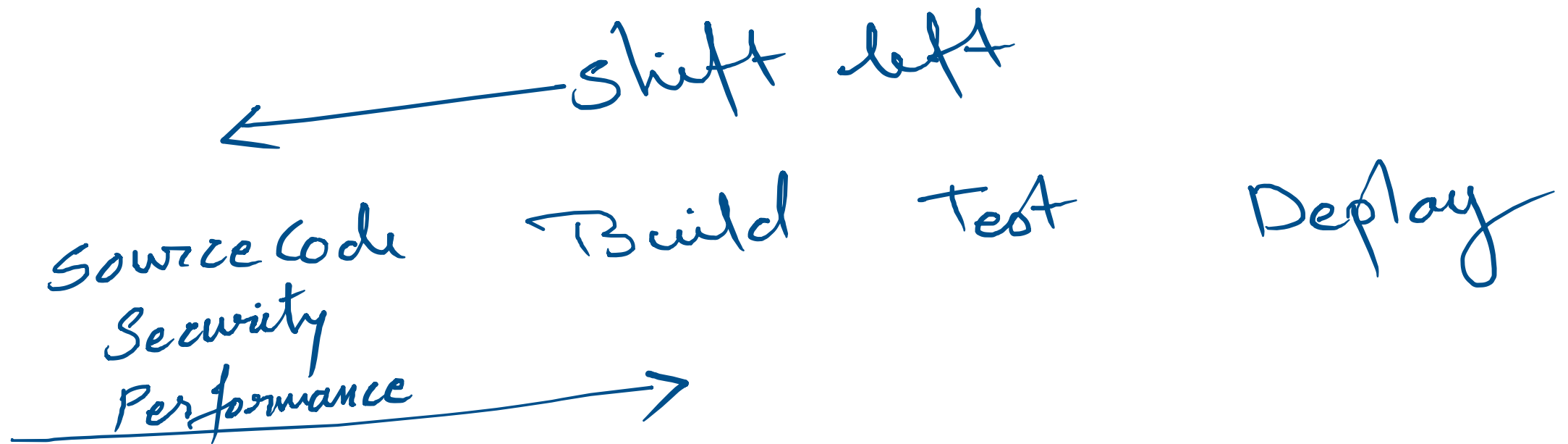






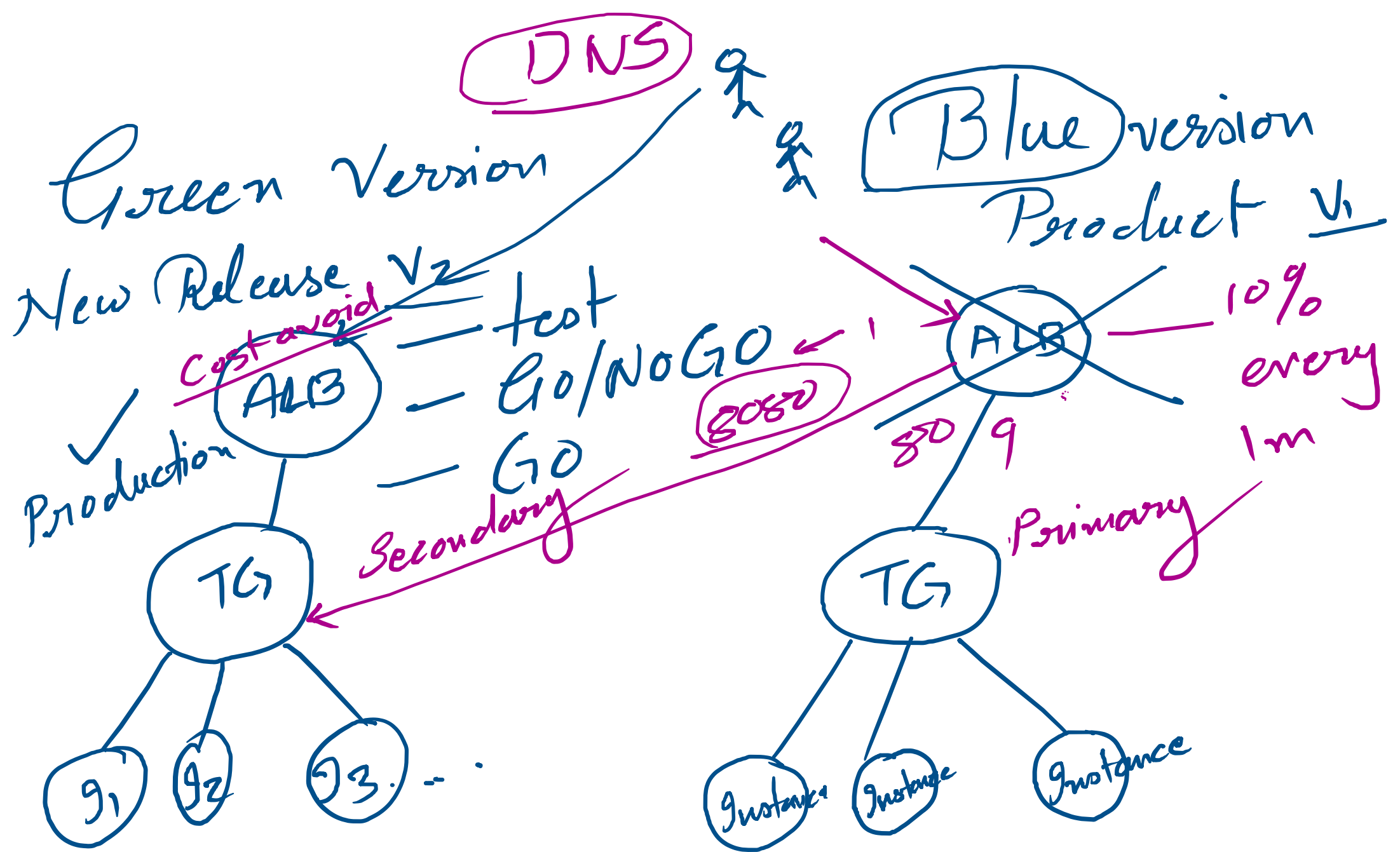
1k8s

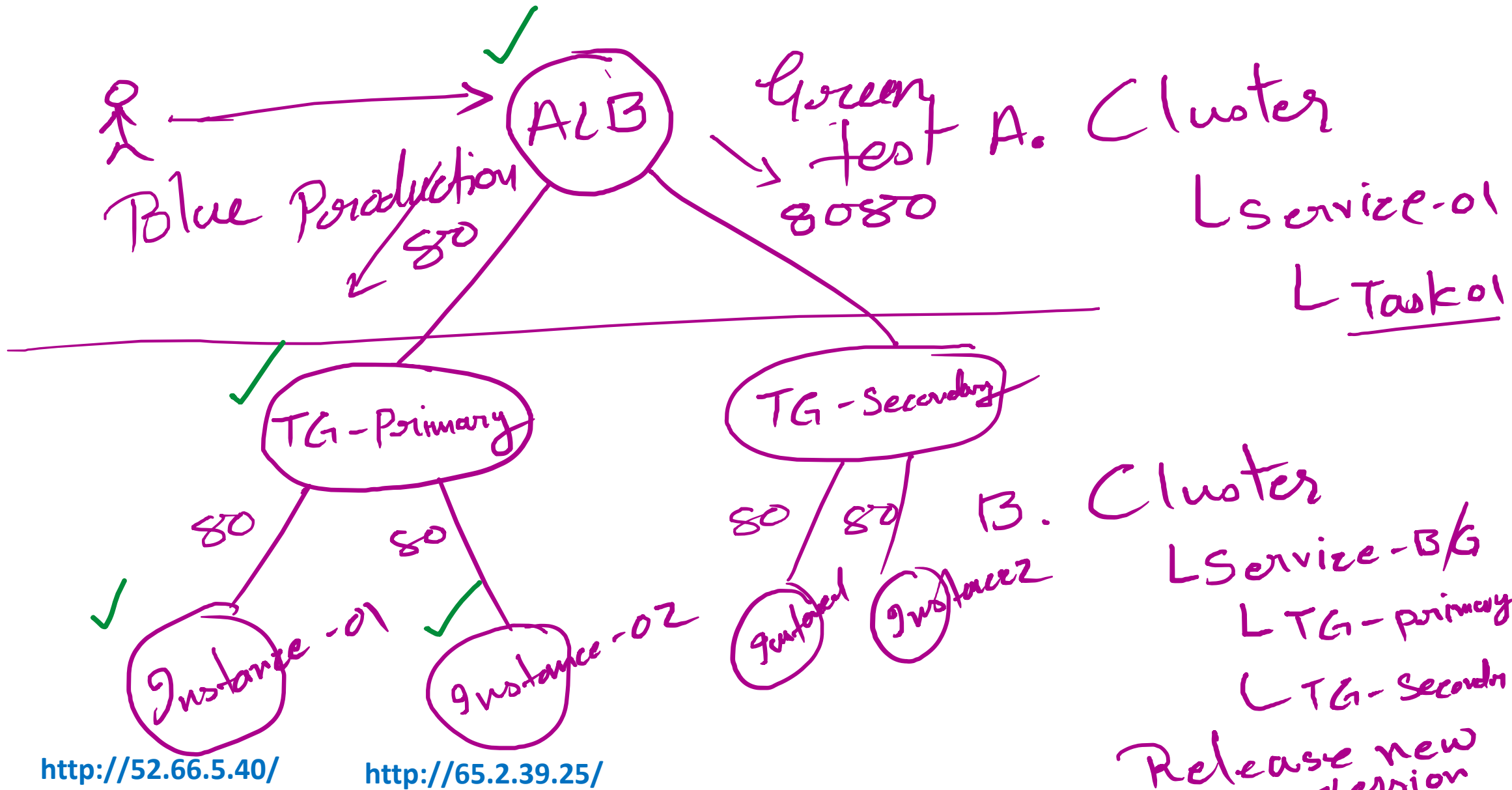




Docker Images

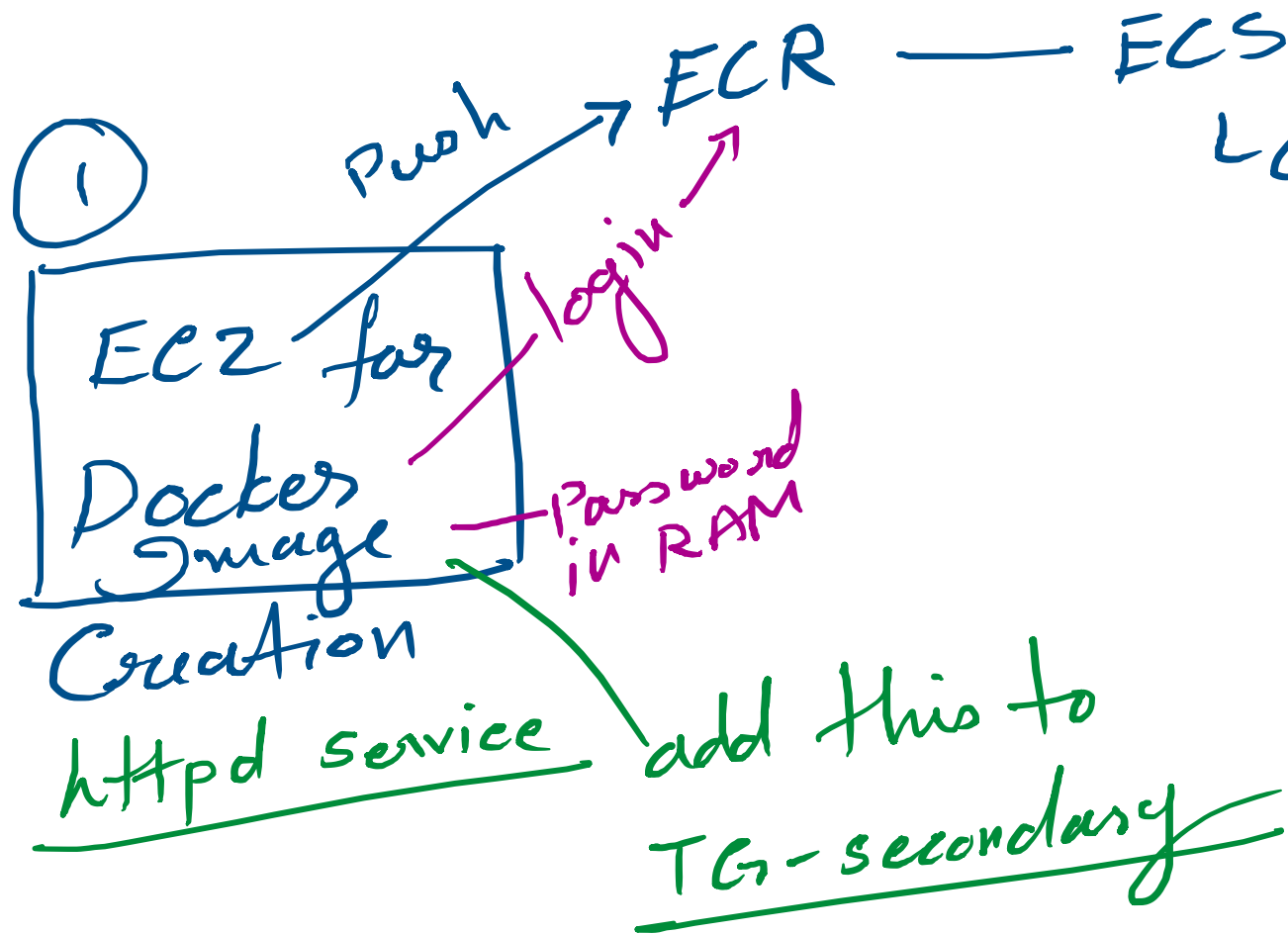
- local machine
- Push image to ECR
- ECS using ECR docker image
- B/G deployment. — ?





<http://52.66.5.40/>

<http://65.2.39.25/>



LC Cluster
Service
Task def

Docker Images

EC2
Deploy Here + 01

Docker Images

EC2
Deploy Here + 01

Fargate

Code Pipeline

Code Commit

Repo for
the application
+ buildspec.yml
+ Dockerfile

Code Build

↓
Create ECR Image
Push to ECR

└─ Automate

- Task Defⁿ
- Service update
 - B/G
 - Roll update

1. Cloud Formation $\Rightarrow$ IaC

2. Serverless ^{- Terraform} serverless

L Lambda

L API Gateway

L Serverless Application

3. IaC for serverless SAM CLI $\Rightarrow$ Cloud Formation

4. Peripheral services

Solution

Infrastructure?

VPC
└ Subnet
└ RT
└ IGW
└ NAT GW

VM m/c

H/W

ALB

EKS



Console - manual
CLI

VM

Layer controlled
aws

IaaS

VM
H/W - BIOS

AWS Cloud

Engine

understand
all services
& how to provision
& release

Console

Manual

Manual

CLI

SDK

CDK

Java

Nugets

Python ...

.Bat

.sh

xyz: {
}

Cloud Formation

Script

JSON

YML

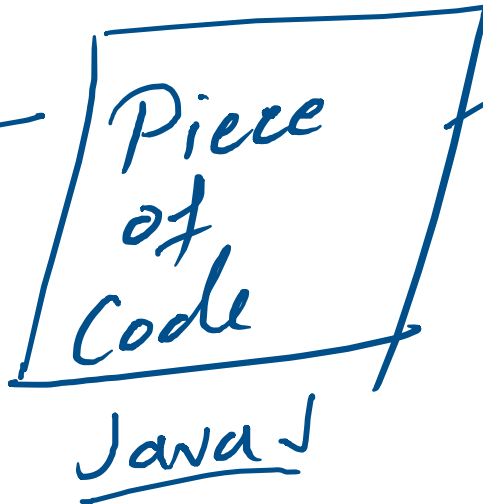
Serverless

└ Underneath Infra is managed by AWS

8✓
11✓

Lambda — Serverless service (VM)

Invoke

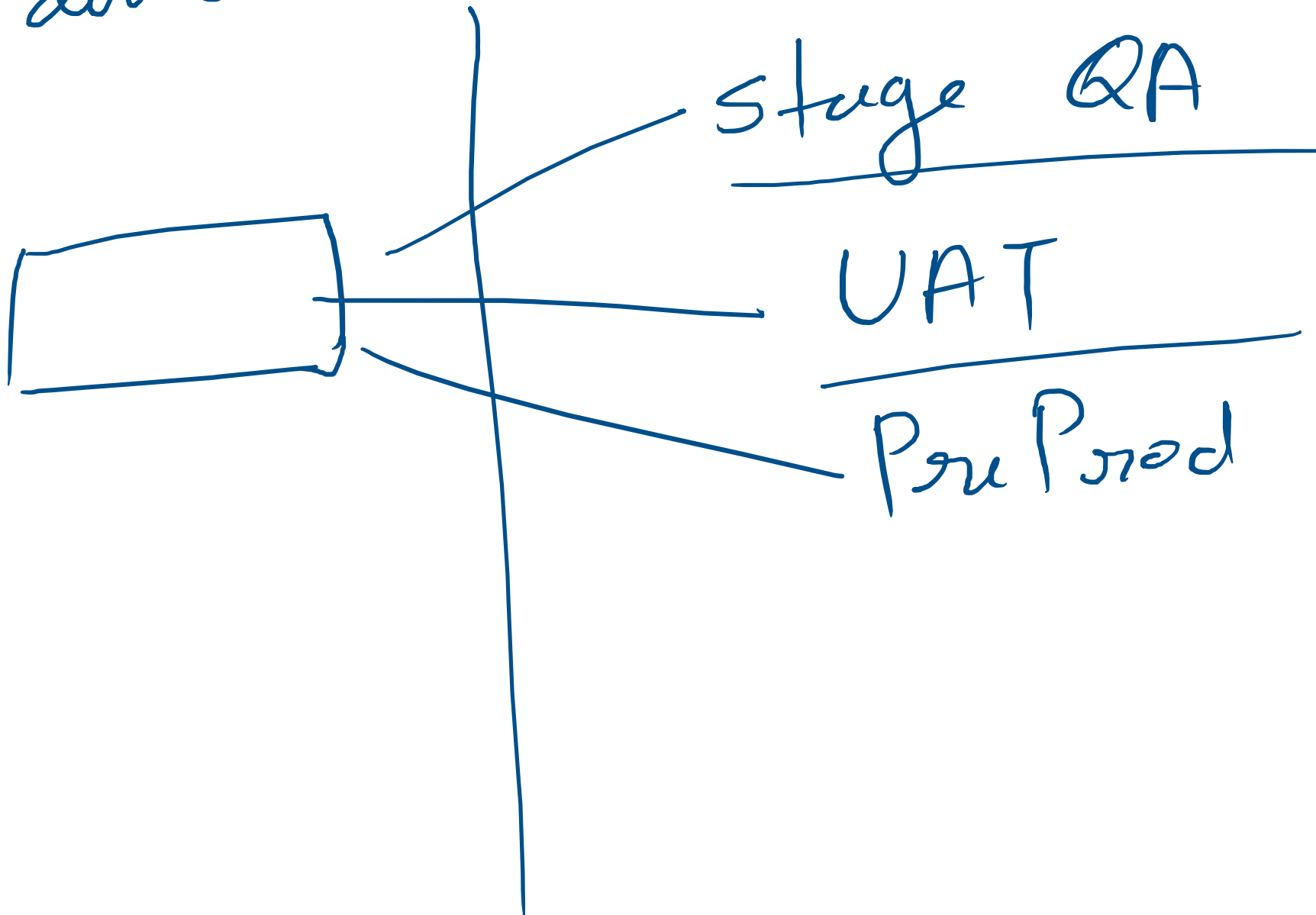


"Hello World!"

|| Compute m/c - H/W
|| OS
|| JVM

100 Java	100 Nodejs
100 Python	100 Go

dev env



API

http

→ Query String

→ headers

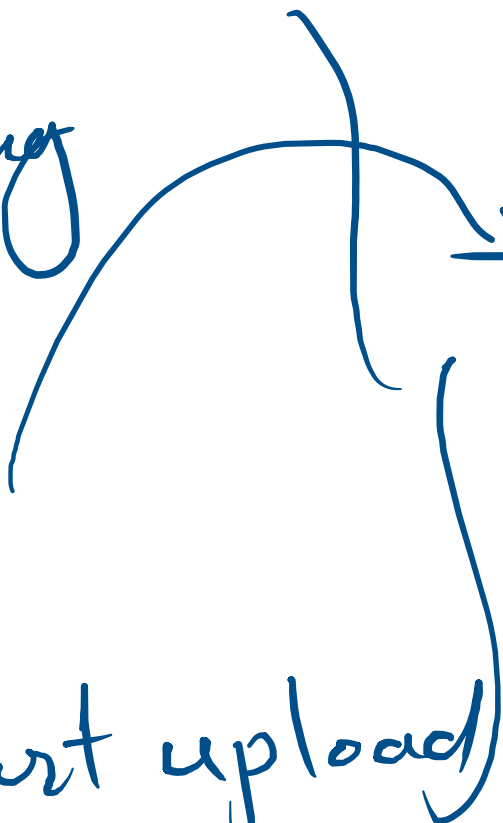
→ Body

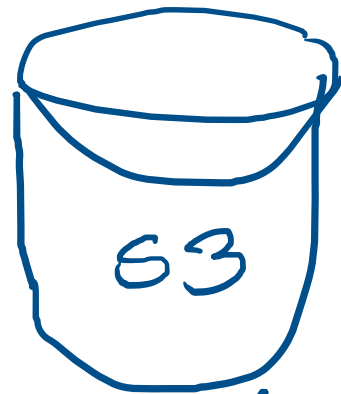
→ Forms

→ multipart upload

Lambda

JSON





Bucket

Store our
notes

Note 1.txt

HTTP

Preflight

Lambda

Role for working
with S3

API Gateway

Role

CORS

Dev

→ endpoint

ARC

web
App

